A

Mini Project

On

AI Eval: AI Driven Evaluation for Enhanced Education Using Machine Learning

(Submitted in partial fulfillment of the requirements for the award of Degree)

BACHELOR OF TECHNOLOGY

In

COMPUTER SCIENCE AND ENGINEERING

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ABSTRACT

The integration of Artificial Intelligence (AI) in education has transformed traditional learning and assessment methods. AI-driven evaluation systems use machine learning (ML) to automate grading, provide personalized feedback, and enhance student learning experiences. These systems can analyze student responses, predict academic performance, and suggest customized learning paths. While AI-driven assessments offer numerous advantages, such as efficiency and scalability, they also present challenges like algorithmic bias and data privacy concerns. This paper explores existing AI-based educational evaluation systems, their advantages and disadvantages, and the necessary hardware and software requirements for implementation.

EXISTING SYSTEMS

- **Accuracy**
- Measures proportion of correctly classified instances.
- Evaluates overall model performance.
- Provides a general idea of model accuracy.

- **Precision**

  - Measures true positives among all positive predictions.
  - Evaluates model's ability to avoid false positives.
  - Provides insight into model's precision.

- **Recall**

  - Measures true positives among all actual positive instances.
  - Evaluates model's ability to detect true positives.
  - Provides insight into model's recall.

- **Unsupervised Learning**

  - Trains models on unlabeled data.
  - Discovers hidden patterns and relationships.
  - Enables clustering, dimensionality reduction, and anomaly detection.

These components provide a solid foundation for the project. The next steps would be to enhance and expand on these existing components to further improve the system's performance and capabilities.

DISADVANTAGES

- **Lack of Human Judgment** – AI cannot fully understand creative or emotional aspects of student responses.
- **Privacy Concerns** – Student data security and compliance with regulations (GDPR, FERPA).
- **Dependence on Internet & Technology** – Requires strong infrastructure, limiting access in rural areas.

PROPOSED SYSTEMS

➢ Data Plagiarism Detection

- Implement a plagiarism detection tool that allows users to make changes to their submissions with a 40% similarity threshold.
- This feature will encourage original work while providing flexibility for users to build upon existing ideas.

➢ Reference File Submission

- Allow users to submit reference files to the bot, making it easier for the bot to evaluate the submission.
- This feature will enable the bot to better understand the context and content of the submission.

➢ Efficiency Rate Evaluation

- Develop an efficiency rate metric to evaluate the effectiveness of the evaluation process.
- This metric will help identify areas for improvement and optimize the evaluation process.

➢ Enhanced Data Security

- Implement robust security measures to protect the data, including encryption, access controls, and secure storage.
- This feature will safeguard the integrity and confidentiality of the data, ensuring a secure evaluation process.

ADVANTAGES

- **Automated Grading** – Saves educators' time and ensures consistency.
- **Personalized Learning** – Adapts to student strengths and weaknesses.
- **Real-Time Feedback** – Improves student engagement and comprehension.
- **Scalability** – Can handle large volumes of students efficiently.

SYSTEM REQUIREMENTS

➤ HARDWARE REQUIREMENTS:

Component	Specification
Processor	Intel i7 / AMD Ryzen 7 or higher
RAM	Minimum 16GB (Recommended 32GB for large datasets)
Storage	SSD (512GB or more) for fast processing
GPU	NVIDIA RTX 3060+ (for AI model training)
Internet Connection	High-speed broadband (for cloud-based solutions)
Camera & Microphone	Required for proctoring systems

➤ SOFTWARE REQUIREMENTS:

Component	Tools / Frameworks
Programming Languages	Python, R
AI & ML Libraries	TensorFlow, PyTorch, Scikit-learn
Database	MySQL, PostgreSQL, MongoDB
Cloud Services	AWS, Google Cloud, Microsoft Azure
NLP Frameworks	spaCy, NLTK, OpenAI GPT models
Security	SSL encryption, GDPR-compliant frameworks